

Linear Algebra

Elementary Row Operations

MPS Department|FEU Institute of Technology



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OBJECTIVES

At the end of the discussion, the learner should be able to:

- Define and enumerate elementary row operations
- Perform elementary row operations
- Identify which matrix is in row-echelon form or reduced-row-echelon form
- Transform matrix into row-echelon form using Gaussian and
- Transform matrix into reduced-row-echelon form using Gauss-Jordan Reduction methods



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Elementary Row Operations

The following are elementary row operations:

1. Interchange two rows.
2. Multiply a row by nonzero constants.
3. Add a multiple of a row to another row.

These three operations also correspond to **elementary column operations**. You just need to change the word row by column in each operations.

Elementary Row Operations

Example: Given matrix $A = \begin{bmatrix} -2 & 1 & -1 & 2 \\ 1 & -2 & 3 & 0 \\ 3 & 1 & -1 & 2 \end{bmatrix}$, perform

$2R_2 + R_1 \rightarrow R_1$. i.e., 2 times row 2 plus row 1 then change row 1.

Answer: $\begin{bmatrix} 0 & -3 & 5 & 2 \\ 1 & -2 & 3 & 0 \\ 3 & 1 & -1 & 2 \end{bmatrix}$

Example

Perform the indicated elementary row operations in the given matrices.

1.
$$\begin{bmatrix} 10 & 10 \\ -5 & 5 \\ 15 & 10 \\ 10 & -5 \end{bmatrix} (R_2 - 2R_4 \rightarrow R_4) \text{ change the entries in } R_4 \text{ by } R_2 \text{ minus (2 times } R_4)$$

2.
$$\begin{bmatrix} 0 & -1 & 5 & 1 & 0 \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix} (R_3 + 3R_1 \rightarrow R_1) \text{ change the entries in } R_1 \text{ by } R_3 \text{ plus (3 times } R_1)$$

3.
$$\begin{bmatrix} 2 & -9 & 11 \\ -3 & 11 & 6 \\ -8 & 5 & -7 \end{bmatrix} (R_1 \leftrightarrow R_3) \text{ interchange row 1 and row 3.}$$

Example

Perform the indicated elementary row operations in the given matrices.

$$1. \begin{bmatrix} 10 & 10 \\ -5 & 5 \\ 15 & 10 \\ 10 & -5 \end{bmatrix} (R_2 - 2R_4 \rightarrow R_2) \begin{bmatrix} 10 & 10 \\ -5 & 5 \\ 15 & 10 \\ -5 - 2(10) & 5 - 2(-5) \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ -5 & 5 \\ 15 & 10 \\ -25 & 15 \end{bmatrix}$$

$$2. \begin{bmatrix} 0 & -1 & 5 & 1 & 0 \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix} (R_3 + 3R_1 \rightarrow R_3)$$

$$\begin{bmatrix} 1 + 3(0) & -1 + 3(-1) & 5 + 3(5) & 1 + 3(1) & 0 + 3(0) \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 14 & 4 & 8 \\ 4 & 2 & 3 & -4 & 4 \\ 1 & 3 & -1 & 1 & 8 \end{bmatrix}$$

$$3. \begin{bmatrix} 2 & -9 & 11 \\ -3 & 11 & 6 \\ -8 & 5 & -7 \end{bmatrix} (R_1 \leftrightarrow R_3) = \begin{bmatrix} -8 & 5 & -7 \\ -3 & 11 & 6 \\ 2 & -9 & 11 \end{bmatrix}$$

Example

Perform the indicated elementary row operations in the given matrices.

1. $\begin{bmatrix} 1 & 9 \\ -10 & 15 \\ 5 & 10 \\ 10 & 2 \end{bmatrix} \left(\frac{1}{5} R_2 \right)$ change the entries in R_2 by multiplying $\frac{1}{5}$ in each entries in R_2

2. $\begin{bmatrix} 3 & 2 \\ -4 & 1 \end{bmatrix} (3R_1 + 2R_2 \rightarrow R_2)$ change the entries in R_2 by (3 times R_1) plus (2 times R_2)

3. $\begin{bmatrix} 4 & -1 & 4 \\ -3 & 1 & 6 \\ 8 & 3 & -1 \end{bmatrix} (R_1 \leftrightarrow R_2)$ interchange row 1 and row 2.

Example

Perform the indicated elementary row operations in the given matrices.

$$1. \begin{bmatrix} 1 & 9 \\ -10 & 15 \\ 5 & 10 \\ 10 & 2 \end{bmatrix} \left(\frac{1}{5} R_2 \right) \begin{bmatrix} 1 & 9 \\ \left(\frac{1}{5}\right)(-10) & \left(\frac{1}{5}\right)(15) \\ 5 & 10 \\ 10 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ -2 & 3 \\ 5 & 10 \\ 10 & 2 \end{bmatrix}$$

$$2. \begin{bmatrix} 3 & 2 \\ -4 & 1 \end{bmatrix} (2R_2 + 3R_1 \rightarrow R_1) \begin{bmatrix} 2(-4) + 3(3) & 2(1) + 3(1) \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ -4 & 1 \end{bmatrix}$$

$$3. \begin{bmatrix} 4 & -1 & 4 \\ -3 & 1 & 6 \\ 8 & 3 & -1 \end{bmatrix} (R_1 \leftrightarrow R_2) \begin{bmatrix} -3 & 1 & 6 \\ 4 & -1 & 4 \\ 8 & 3 & -1 \end{bmatrix}$$

Example

What elementary row operations applied to
$$\begin{bmatrix} -2 & 1 & -1 & 2 \\ 1 & -2 & 3 & 0 \\ 3 & 1 & -1 & 2 \end{bmatrix}$$

will yield the matrix
$$\begin{bmatrix} -2 & 1 & -1 & 2 \\ 1 & -2 & 3 & 0 \\ 0.75 & 0.25 & -0.25 & 0.5 \end{bmatrix} \quad ?$$

Answer: $\frac{1}{4}R_3$

Example

Matrix B is obtained by applying elementary row operation on matrix A . Identify what elementary row operation was used.

$$1. \quad A = \begin{bmatrix} 2 & 3 & 5 \\ 4 & 2 & -6 \\ -1 & 2 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 & 5 \\ -2 & -1 & 3 \\ -1 & 2 & 4 \end{bmatrix}$$

$$2. \quad A = \begin{bmatrix} -2 & 3 & 3 & -1 & 8 \\ 4 & -1 & 5 & 0 & 7 \\ 1 & 3 & -2 & 6 & 9 \\ 9 & 4 & -3 & 4 & -1 \end{bmatrix}, B = \begin{bmatrix} -2 & 3 & 3 & -1 & 8 \\ 4 & -1 & 5 & 0 & 7 \\ -1 & 6 & 5 & 5 & 17 \\ 9 & 4 & -3 & 4 & -1 \end{bmatrix}$$

$$3. \quad A = \begin{bmatrix} 6 & 2 & -6 & 1 & 6 & -1 \\ 4 & -4 & 10 & 0 & 7 & -4 \\ -1 & 6 & 1 & 7 & 5 & 2 \\ -2 & 5 & 0 & 3 & 1 & 5 \\ 3 & -8 & 9 & 5 & 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 6 & 2 & -6 & 1 & 6 & -1 \\ 4 & -4 & 10 & 0 & 7 & -4 \\ -1 & 6 & 1 & 7 & 5 & 2 \\ -2 & 5 & 0 & 3 & 1 & 5 \\ 7 & -12 & 29 & 39 & 16 & 13 \end{bmatrix}$$



Answer

1. $-\frac{1}{2}R_2$

2. $R_1 + R_3 \rightarrow R_3$

3. $2R_3 + 3R_5$



Recall: Elementary Row Operations

The following are elementary row operations:

1. Interchange two rows.
2. Multiply a row by nonzero constants.
3. Add a multiple of a row to another row.

These operations shall be used in transforming a matrix to a row-echelon and reduced row-echelon form.

Row Echelon Form (REF) of a Matrix

A matrix is in **row echelon form** if the following conditions are satisfied:

1. Any rows consisting entirely of zeros occur **at the bottom of the matrix**.
2. For each row that does not consist entirely of zeros, the first nonzero entry is 1 (called **a leading 1**).
3. For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row.



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Example:

The given matrix A below is a matrix in row-echelon form.

$$A = \begin{bmatrix} 1 & -2 & 1 & 7 \\ 0 & 1 & -2 & -7 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

All rows consist of nonzero entries. Also For each row that does not consist entirely of zeros, the first nonzero entry is 1. For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row. Thus, matrix A is in row-echelon form.

Example

The following matrices are in row echelon form:

$$\begin{bmatrix} 1 & -1 & 5 & 1 & 0 \\ 0 & 1 & 3 & -4 & 4 \\ 0 & 0 & 1 & 0 & 8 \end{bmatrix}, \quad \begin{bmatrix} 1 & -9 & 0 \\ 0 & 1 & 6 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 3 & -1 & 8 \\ 0 & 1 & 5 & 0 & 7 \\ 0 & 0 & 0 & 1 & 17 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



German mathematician **Carl Friedrich Gauss** is recognized, with Newton and Archimedes, as one of the three greatest mathematicians in history. Gauss used a form of what is now known as Gaussian elimination in his research. Although this method was named in his honor, the Chinese used an almost identical method some 2000 years prior to Gauss.



Source: **Ron Larson, *Linear Algebra 8th edition***

Transforming into Row-Echelon Form

The process of transforming a matrix into row-echelon form is called **Gaussian Elimination**. This named after the German mathematician **Carl Friedrich Gauss**.

Example

Find a row echelon form for the given matrix.

$$\begin{bmatrix} 1 & 2 & -3 \\ -3 & -6 & 10 \\ -2 & -4 & 7 \end{bmatrix}$$

Solution:

Since there are no rows consisting entirely zero entries and the first entry in R_1 is 1, then we now transform the matrix such that the entries below a_{11} are all zeros.

$$\begin{bmatrix} 1 & 2 & -3 \\ -3 & -6 & 10 \\ -2 & -4 & 7 \end{bmatrix} \xrightarrow{3R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ -2 & -4 & 7 \end{bmatrix}$$

$$\xrightarrow{2R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-1R_2 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Finally, we have transformed the matrix into row-echelon form

Here, note that the leading entry in R_2 is in a_{23} . Thus, we need to transform the obtained matrix so that the entry below a_{23} is 0.

Example:

Transform the given matrix into row-echelon form.

$$\begin{bmatrix} 3 & 5 & 4 \\ -2 & -3 & 2 \end{bmatrix}$$

Solution:

This matrix does not contain a row with entirely zero entries. Looking at the entries in R_1 , all of them are divisible by 3 and dividing it by 3 will give us $a_{11} = 1$.

i.e.,

$$\begin{bmatrix} 3 & 21 & 42 \\ -2 & -3 & 2 \end{bmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{bmatrix} 1 & 7 & 14 \\ -2 & -3 & 2 \end{bmatrix}$$

Next, the entry below the leading entry $a_{11} = 1$ should be zero. To obtain this, we need to multiply R_1 by 2 and add it to R_2 . The result should be used to replace the entries in R_2 .

Continuation

$$\begin{bmatrix} 1 & 7 & 14 \\ -2 & -3 & 2 \end{bmatrix} \xrightarrow{2R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 7 & 14 \\ 0 & 11 & 30 \end{bmatrix}$$

Now it only remains to transform the matrix such that the entries in the 1st column is untouched but the leading entry in the 2nd row should be 1. To do this, we need to divide R_2 by 11.

$$\begin{bmatrix} 1 & 7 & 14 \\ 0 & 11 & 30 \end{bmatrix} \xrightarrow{\frac{1}{11}R_2} \begin{bmatrix} 1 & 7 & 14 \\ 0 & 1 & \frac{30}{11} \end{bmatrix}.$$

We have obtained our desired form!

Transforming into Row-Echelon Form

Example: Transform the given matrix into REF

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 2 & 1 & -1 & 0 \\ -1 & 2 & -3 & 7 \end{bmatrix}.$$

-2 times row 1 plus row 2,
then replace row 2

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 2 & 1 & -1 & 0 \\ -1 & 2 & -3 & 7 \end{bmatrix} \xrightarrow{-2R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 3 & -5 & 6 \\ -1 & 2 & -3 & 7 \end{bmatrix}$$

Row 1 plus row 3 change row 3

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 3 & -5 & 6 \\ 0 & 1 & -1 & 4 \end{bmatrix} \xrightarrow{R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 3 & -5 & 6 \\ 0 & 1 & -1 & 4 \end{bmatrix}$$

Interchange row 2 and row 3

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -1 & 4 \\ 0 & 3 & -5 & 6 \end{bmatrix}$$



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Transforming into Row-Echelon Form

Continuation:

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -1 & 4 \\ 0 & 3 & -5 & 6 \end{bmatrix} \xrightarrow[-3R_2 + R_3 \rightarrow R_3]{\substack{\text{-3 times row 1 plus row 3,} \\ \text{then replace row 3}}} \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & -2 & -6 \end{bmatrix}$$

Multiply row 3 by $-\frac{1}{2}$

$$\xrightarrow{-\frac{1}{2}R_3}$$

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

This is a row-
echelon form
of the given
matrix



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Reduced Row Echelon Form (RREF) of a Matrix

A matrix in row-echelon form is in **reduced row-echelon form** when every column that has a leading 1 has zeros in every position **above and below its leading 1**.



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Example

Below are some examples of matrices in RREF.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Transforming into Reduced Row-Echelon Form

In Gaussian elimination, we apply elementary row operations to a matrix to obtain a row-echelon form. A second method of elimination is called **Gauss-Jordan elimination** after **Carl Friedrich Gauss** and **Wilhelm Jordan**. This continues the reduction process until a *reduced row-echelon form* is obtained.

Transforming into Reduced Row-Echelon Form

Example: Transform the given matrix into RREF $A = \begin{bmatrix} 3 & 7 & -11 & 44 \\ 1 & 2 & -3 & 12 \\ 4 & 9 & -13 & 53 \end{bmatrix}$.

$$\xrightarrow{-3R_2 + R_1 \rightarrow R_1} \begin{bmatrix} 0 & 1 & -2 & 8 \\ 1 & 2 & -3 & 12 \\ 4 & 9 & -13 & 53 \end{bmatrix} \xrightarrow{-4R_2 + R_3 \rightarrow R_3} \begin{bmatrix} 0 & 1 & -2 & 8 \\ 1 & 2 & -3 & 12 \\ 0 & 1 & -1 & 5 \end{bmatrix}$$

$$\xrightarrow{-R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 0 & 1 & -2 & 8 \\ 1 & 2 & -3 & 12 \\ 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & -3 & 12 \\ 0 & 1 & -2 & 8 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$



Transforming into Reduced Row-Echelon Form

$$\begin{bmatrix} 1 & 2 & -3 & 12 \\ 0 & 1 & -2 & 8 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

Note that, this matrix is now a row-echelon form.

To obtain a reduced row-echelon form, we need to continue applying elementary row operations to our REF matrix until every column that has a leading 1 has 0's elsewhere.

Transforming into Reduced Row-Echelon Form

Continuing the process, we have

$$\begin{bmatrix} 1 & 2 & -3 & 12 \\ 0 & 1 & -2 & 8 \\ 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow{-2R_2 + R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 1 & -4 \\ 0 & 1 & -2 & 8 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

$$\xrightarrow{-R_3 + R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -2 & 8 \\ 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow{2R_3 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

The reduce row-
echelon form of
matrix A

Transforming into Reduced Row-Echelon Form

Example: This example shows a matrix A together with its reduced-row-echelon form which can be verified by applying finite series of elementary row operations.

$$A = \begin{bmatrix} 2 & -1 & 1 & -3 & 5 \\ 1 & 2 & 3 & -1 & 16 \\ 7 & -3 & -2 & 2 & -11 \\ -1 & 1 & -1 & 5 & -9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix}$$

reduced-row-echelon
form of A



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Transforming into Reduced Row-Echelon Form

Example: This example shows a matrix B together with its reduced-row-echelon form which can be verified by applying finite series of elementary row operations.

$$B = \begin{bmatrix} 1 & 2 & -3 & 0 & -1 \\ 2 & 3 & -4 & 1 & -1 \\ 3 & 5 & -7 & 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

reduced-row-echelon
form of A

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